

Guiding Visually Challenged Using Raspberry Pi



GURUNATH REDDY M.

Blindness is a state of lacking the visual perception due to physiological or neurological factors. Partial blindness represents the lack of integration in growth of the optic nerve or visual centre of the eye, and total blindness is full absence of the visual light perception. In this project, a simple, cheap and user-friendly smart blind guidance system is designed and implemented to improve the mobility of both blind and visually impaired people in a specific area.

The project includes wearable equipment comprising earphones connected to Raspberry Pi and a hand-stick containing infrared sensors circuit to help the blind person navigate alone safely and avoid any obstacles. The reflected infrared signals from an obstacle are used as inputs to Raspberry Pi. Based on the direction of the received signal, Raspberry Pi prompts a message in

the earphones to help take suitable action.

Circuit and working

Fig. 1 shows the pin description of a GPIO connector on Raspberry Pi. The GPIO connector has pins with different functions. The device has seven true GPIO pins, I²C interface, SPI interface, serial T_X/R_X pins and PWM pins that can be used to control external hardware.

Fig. 2 shows the circuit diagram of the sensor section. It comprises quad operational amplifier LM324 (IC1), three IR transmitter-receiver pairs and

a few easily available components. The sensor section is connected to the GPIO pins of Raspberry Pi as shown in Fig. 2. This section gets the power for operation from Raspberry Pi itself.

The IR transmitter-receiver pairs are mounted on the left, front and right side of the hand stick and they transmit IR signals regularly. The corresponding receiver receives back the signal if it is reflected by some obstacle in that particular direction. The received signal is fed to corresponding operational amplifier of IC1. Output of each operational amplifier is limited to 3.3V by zener diodes. The Raspberry Pi detects the direction from which the signal is received and issues suitable audio message to the earphones to help avoid the obstacle. The accompanied table mentions all the messages. Presets VR1 through VR3 are used to set the reference levels for comparators A1, A4 and A3.

The person who is wearing the earphones gets the suitable commands to take necessary action. One such example is, if there is obstacle in all the directions, the user gets a message 'better to go back' through the earphones connected to the Raspberry Pi.

Software

Before you follow the software installations, ensure that your Raspberry Pi is already setup with Raspbian 'wheezy' operating system. You can refer 'Getting Started with Raspberry Pi' published in April 2013 issue to set up the Raspberry Pi. Now all you need is a network connection on Raspberry Pi to

Different Situations and Messages Given by the System for the User to Take Necessary Actions

Levels at pins 11, 12 and 13 of Raspberry Pi	Message generated by the system for necessary actions
000	You can move in any direction
001	Move either left or front
010	Move either left or right
011	Move left
100	Move either front or right
101	Move front
110	Move right
111	Better to go back

3.3V	1	2	3V
I2CO SDA	3	4	DNC
I2CO SCL	5	6	GROUND
GPIO4	7	8	UART TXD
DNC	9	10	UART RXD
GPIO 17	11	12	GPIO 18
GPIO 21	13	14	DNC
GPIO 22	15	16	GPIO 23
DNC	17	18	GPIO 24
SPI0 MOSI	19	20	DNC
SPI0 MISO	21	22	GPIO 25
SPI0 SCLK	23	24	SPI0 CE0 N
DNC	25	26	SPI0 CE1 N

Fig. 1: Pin description of GPIO header

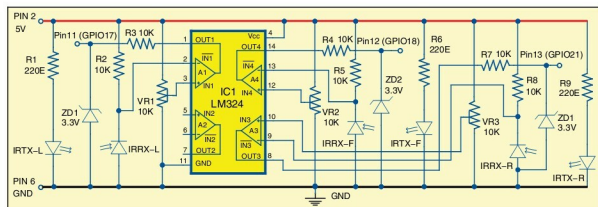


Fig. 2: Circuit diagram of the sensor section